

Functional Requirements Document

Daastaan-e-Pakistan

Nationwide Tourism Marketplace Connecting Travellers with Local Guides

Tagline: *Discover Pakistan. Live Its Stories.*

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Document Control

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Engagement Type	End-to-end platform development (web application, admin panel, backend API, mobile API contract)

Document History

Version	Date	Prepared By	Reviewed By	Approved By	Summary
v0.1	19 Oct 2026	Ahmad Fawad, Flyxosoft	*Pending*	*Pending*	Initial FRD draft translating BRD business requirements (BR-001 to BR-050) into functional specifications per actor, module, and screen.

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Related Documents

Document	File	Status
Business Requirements Document (BRD)	`docs/01-BRD.md` / `.pdf` v0.2	Draft
Software Requirements Specification (SRS)	`docs/03-SRS.md`	*To be produced*
UML Diagrams	`docs/uml/`	*To be produced*
Work Breakdown Structure (WBS)	`docs/04-WBS.md`	*To be produced*

Document Status

Current Status	🟡 **Draft — Awaiting Client Review**
Pending Client Inputs	Lahore activity catalog (drives package tier content), guide roster, final tier naming
Target Launch (v1)	mid-January 2027 (approximately 12 weeks from build commencement)

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Part I — Introduction

1. Purpose

This Functional Requirements Document (FRD) describes, in detail, every functional capability the Daastaan-e-Pakistan platform must deliver at v1 launch. It translates the business-level requirements in the BRD (BR-001 to BR-050) into actor-level, screen-level, interaction-level specifications that can be implemented, tested, and signed off.

Each functional requirement is numbered `FR-###` and tagged with the BRD requirement it fulfils, enabling full traceability.

2. Scope

2.1 In Scope (v1)

- Customer-facing web application (Next.js) **built as a Progressive Web App (PWA)**

- Admin panel (Next.js, protected)
- Backend REST API (NestJS, serving web + admin in v1; mobile in Phase 2)
- OpenAPI contract (used by the Phase 2 Flutter mobile developer)
- **Multi-city catalog** — cities as first-class entities, dedicated per-city SEO landing pages
- **Multi-day packages** — itinerary spans multiple days; booking flow handles date ranges
- **Group bookings** — party-size selector (1–12 at launch) with group pricing rules
- Universal safety features via PWA (SOS, Web Geolocation live tracking, guide check-ins)
- Web Push notifications (via service worker)
- All features described in BRD §8.1 **In Scope (v1)**

2.2 Deferred to Phase 2 (post-v1 launch, weeks 13–20)

- **Native Flutter mobile app** — iOS + Android, customer + guide experiences
- Native push notifications (FCM)
- Advanced offline mode (full offline check-ins with background sync)

2.3 Deferred to v2+

- Cities beyond v1 launch set (added incrementally)
- Group bookings beyond 12 people
- Guide self-serve onboarding portal
- In-app chat (v1 uses WhatsApp click-to-chat)
- Urdu localisation
- Any additional items in BRD §8.2

2.4 PWA Rationale

In v1, mobile users install the web app to their home-screen (PWA). This delivers an app-like experience without the App Store distribution overhead:

- Service worker caches the catalog + bookings for offline viewing
- Web Geolocation API provides GPS for live tracking + SOS coordinates
- Web Push API delivers notifications
- Add-to-home-screen manifest provides launcher icon + splash
- Fulfils v1 safety features without a native mobile build

The Phase 2 Flutter app replaces/augments this with native capabilities (background tracking, camera-first review flow, native push via FCM, deeper OS integration).

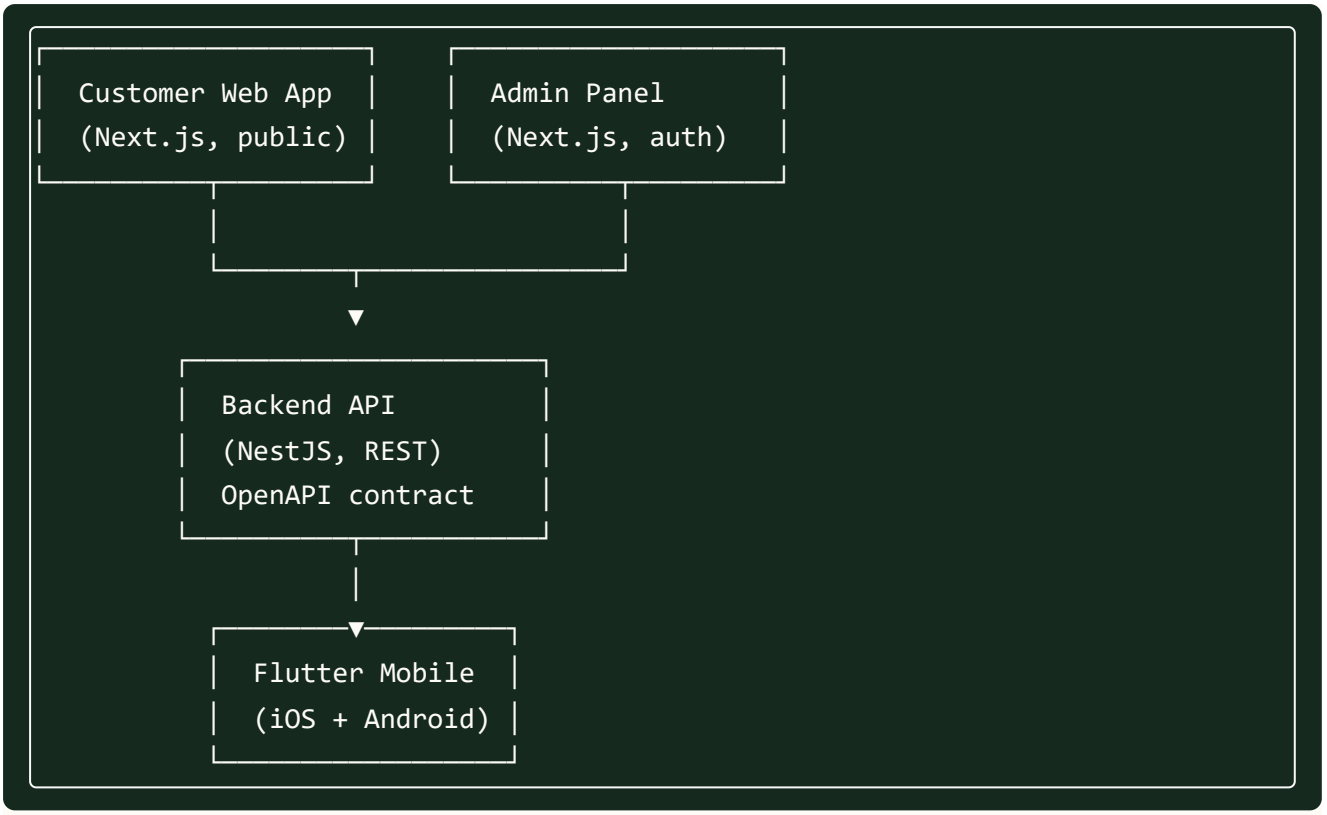
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3. Definitions & Acronyms

Term	Definition
FR-###	A functional requirement defined in this document
BR-###	A business requirement defined in the BRD
Actor	A role interacting with the system (e.g. Customer, Guide, Admin)
Tier	A package category (Cultural Heritage, Food Trail, Family, Nature)
Slot	A specific date + time window when a package can be booked (e.g. 2026-11-14 09:00–15:00)
Booking	A confirmed, paid reservation of a package by a customer on a slot with a specific guide
Itinerary	The time-ordered plan of activities for a booking
SOS	Emergency alert initiated by the customer during an active trip
RBAC	Role-Based Access Control
CMS	Content Management System (admin-side content editor)
CTA	Call-to-action (button)
CNIC	Computerised National Identity Card (Pakistan)

4. System Overview

Daastaan-e-Pakistan is composed of four logical components, all backed by a single API:



All actors interact with the API; no actor talks directly to the database.

5. Actors & Roles

Actor	Access Surface	Primary Capabilities
Guest Visitor	Customer Web (public)	Browse packages, read reviews, subscribe to newsletter, read content pages
Registered Customer	Customer Web + Mobile	Everything above + book, pay, review, refer, use SOS (Women-First)
Guide	Mobile App (guide view)	View assigned bookings, perform trip check-ins, update availability, view earnings
Super Admin	Admin Panel	Full access to every admin feature
Ops Admin	Admin Panel	Manage guides, packages, bookings, payments, reviews — cannot edit RBAC or billing settings
Content Manager	Admin Panel	Manage blog, travel guides, FAQ, static content, newsletter — no booking/payment access

RBAC enforcement is specified in §15.

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Part II — Customer Web Application

6. Discovery & Public Browsing

6.1 Overview

The customer web app is the primary acquisition surface. Non-authenticated visitors must be able to discover the brand, understand the offering, browse the full catalog, and take low-commitment actions (newsletter sign-up, click-to-share) before deciding to book.

6.2 Related BRs

BR-001, BR-002, BR-003, BR-004, BR-005, BR-006, BR-007

6.3 Landing Page

FR ID	Requirement
FR-001	The landing page shall load within 2.5 seconds (LCP) on 4G.
FR-002	The hero section shall display a brand tagline, a short value proposition, and a primary CTA ("Explore Tours").
FR-003	The hero section shall feature a background image or video representing authentic Pakistan — editable by admin.
FR-004	A **"How It Works"** section shall display four steps: *Discover → Book → Experience → Share*.
FR-005	A **"Package Categories"** section shall display tier cards (Cultural Heritage, Food Trail, Family, Nature) — each clickable to a filtered catalog view.
FR-006	A **"Featured Packages"** carousel shall display 6–12 packages, auto-selected by admin flag or booking volume.
FR-007	A **"Trusted Guides"** strip shall display 4–8 verified guide avatars (name, speciality, rating).
FR-008	A **"Latest Reviews"** section shall show the 5 most recent approved reviews (rating, text excerpt, package name).
FR-009	A **"From the Blog"** strip shall display the 3 most recent published blog posts (title, cover image, read time).
FR-010	A footer newsletter capture shall accept an email and submit without page reload.
FR-011	The footer shall include links to About, Vision, Mission, Our Story, Blog, Travel Guide, FAQ, Safety Guide, Contact, Terms, Privacy, Social handles.

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6.4 Catalog Browsing

FR ID	Requirement
FR-012	A city picker in the header shall let the visitor switch between v1 cities (Lahore, Islamabad, and any others at launch). Current city persists in `localStorage`.
FR-012a	Each city shall have a dedicated SEO landing page at `/explore/[city]` with city-specific hero image, intro copy, package list, and local reviews.
FR-012b	The master catalog page `/explore` shall list all active packages across all cities, with a city filter chip.
FR-013	A filter sidebar shall allow filtering by City, Tier, Duration (hours OR days), Group Size, Language, Guide Rating.
FR-014	A sort control shall allow sorting by Popular (default), Highest Rated, Newest.
FR-015	Each package card shall display: cover image, package name, tier badge, duration, star rating, review count, "View Details" CTA.
FR-016	**Price shall NOT be surfaced on catalog cards** (per client decision — pricing is revealed on detail page after context).
FR-017	The catalog shall paginate (12 per page) or infinite-scroll.
FR-018	The URL shall reflect filters as query params (shareable).
FR-019	A guest shall be able to view the catalog without authentication.

6.5 Content Discovery

FR ID	Requirement
FR-020	A `/blog` route shall list published blog posts (card grid), filterable by category/tag.
FR-021	A `/travel-guide` route shall list travel guide articles (packing lists, cultural tips, city guides).
FR-022	Each blog post shall support rich content (headings, images, embedded videos, quotes, lists).
FR-023	Each content page shall carry Open Graph + Twitter card meta tags for social preview.
FR-024	A `/faq` route shall display categorised FAQ accordions.

6.6 Newsletter Subscription

FR ID	Requirement
FR-025	A visitor shall subscribe by submitting an email address (validated format).
FR-026	On submission, the system shall confirm success with a toast message and send a welcome email.
FR-027	Duplicate emails shall be silently accepted (no leakage of subscription status).
FR-028	A GDPR-compliant consent checkbox shall be present for EU IPs (shown via geolocation).
FR-029	An **unsubscribe link** shall be included in every newsletter email.

7. Authentication & Account Management

7.1 Overview

Guests become customers to book. Signup is low-friction, offers multiple methods, and preserves intent (returning user to the booking flow they were on).

7.2 Related BRs

BR-008, BR-009, BR-010

7.3 Signup

FR ID	Requirement
FR-030	The customer shall sign up using any of: email + password, Google OAuth, Facebook OAuth, Apple Sign-In (mobile only in v1).
FR-031	Signup form (email) shall capture: full name, email, password (min 8 chars, 1 upper, 1 digit), accept T&C.
FR-032	On email signup, the system shall send a verification email with a 24-hour link; booking is permitted while unverified but flagged.
FR-033	On social signup, email verification is skipped (relied on by the provider).
FR-034	After signup, the user is logged in and redirected to the intended destination (deep-link preserved).
FR-035	The system shall reject passwords commonly breached (haveibeenpwned API optional, offline list acceptable for v1).

7.4 Login

FR ID	Requirement
FR-036	Login shall support email + password and all social providers listed in FR-030.
FR-037	A **"Remember me"** checkbox shall extend session cookie TTL from 24h to 30 days.
FR-038	After 5 failed login attempts within 10 minutes, the account shall be locked for 15 minutes (brute-force protection).
FR-039	A **"Forgot password"** link shall allow password reset via email token (valid 1 hour).

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7.5 Profile Management

FR ID	Requirement
FR-040	The customer shall edit full name, phone, country, profile photo.
FR-041	The customer shall set an **emergency contact** (name, relationship, phone, email) — strongly recommended on first booking; required to activate the universal safety layer.
FR-042	The customer shall change password (requires current password).
FR-043	The customer shall link additional social accounts to one profile.
FR-044	The customer shall delete the account (soft-delete; data retained 30 days then purged except booking records required by law).
FR-045	The system shall log all profile changes to an audit trail (admin-visible).

7.6 Session Handling

FR ID	Requirement
FR-046	Web session shall use HTTP-only, secure, same-site cookies.
FR-047	Mobile shall authenticate via JWT access token (15 min TTL) + refresh token (30 days).
FR-048	On logout, server-side session shall be invalidated; refresh token revoked.

8. Package Detail & Booking Flow

8.1 Overview

The decisive conversion moment. The package detail page must build enough trust and clarity for the customer to proceed to pay. The booking flow is a stepped experience: Slot → Guide → Add-ons → Review → Pay.

8.2 Related BRs

BR-007, BR-010, BR-011, BR-012, BR-013

8.3 Package Detail Page

FR ID	Requirement
FR-049	The URL shall be `/package/[slug]` (SEO-friendly).
FR-050	The page shall display: hero image gallery (3–8 images), package title, tier badge, city, duration (hours or days), group-size range accepted, average rating + review count, short description, "Book Now" CTA.
FR-051	An **"Itinerary"** section shall list activities. For single-day packages, time-ordered within the day. For **multi-day packages**, grouped by day (**"Day 1 — Lahore Arrival & Walled City"**, **"Day 2 — Hunza"** etc.) with each day's activities nested. Includes title, description, image, estimated duration.
FR-052	An **"Included"** section shall list what the package covers (transport, meals, entry fees, concierge for Women-First).
FR-053	An **"Excluded"** section shall list what is NOT covered (personal shopping, tips, etc.).
FR-054	A **"Meet Your Guide"** section shall show the typical guide profile card (name, photo, bio, languages, rating, verified badge). Guide gender shown; customer may filter by preference at booking (FR-065).
FR-055	A **"Reviews"** section shall display approved reviews (pagination or load-more) with filter by rating.
FR-056	A **"Related Packages"** section shall show 3–6 similar packages.
FR-057	A persistent sticky **"Book Now"** button shall appear on mobile scroll.
FR-058	Price shall be displayed on this page (not on catalog cards per FR-016) in the user's currency with FX conversion.

8.4 Booking Step 1 — Date & Group Size

FR ID	Requirement
FR-059	A calendar widget shall display available dates (next 180 days).
FR-060	Unavailable dates (no guide available) shall be visually distinct and non-selectable.
FR-061a	For **single-day packages**, the user picks a single date; available timeslots for that date are listed (start time, end time, duration).
FR-061b	For **multi-day packages**, the user picks a start date; the end date is auto-calculated from `duration_days` ; the system verifies continuous guide availability across the full range.
FR-061c	A **"Group Size"** selector shall let the user choose party size from 1 to the package's `max_party_size` (default cap 12 at launch).
FR-061d	Group pricing shall update in real-time: per-person pricing × party size (with optional group discount tiers set by admin).
FR-062	Slot/date availability is derived from guide availability for the matching tier and city.

8.5 Booking Step 2 — Guide Choice

FR ID	Requirement
FR-063	By default, "We'll assign the best available guide" is pre-selected.
FR-064	A collapsible **"Choose Your Guide"** section shall list guides available for the selected slot (name, photo, rating, speciality, bio, languages, verified badge).
FR-065	An optional **"Guide Preference"** control shall let the customer filter by guide gender (any / female / male) — defaults to "any". No tier forces this filter.
FR-066	The customer shall select one guide or keep the default.

8.6 Booking Step 3 — Add-ons (Universal)

FR ID	Requirement
FR-067	For every package, available optional add-ons shall be listed (hotel pickup, makeover, shopping guidance, gifting pack). Each has a name, description, price, and toggle.
FR-068	The optional **Safety Layer** opt-in shall appear in this step for every booking — presents SOS, live tracking, and emergency contact sharing. Enabled by default when the customer has an emergency contact set.
FR-069	Selected add-ons and safety layer opt-in shall update the running total (and safety status) in real-time.

8.7 Booking Step 4 — Review & Account

FR ID	Requirement
FR-070	A summary panel shall display: package, date, time, guide, add-ons, subtotal, promo discount (if any), total.
FR-071	If the user is not logged in, a login/signup CTA shall block progression (FR-030).
FR-072	If the Safety Layer is opted-in (FR-068) and no emergency contact exists on the profile, an inline form shall collect it before progressing.
FR-073	A **"Pickup Location"** field shall accept hotel name/address — required if the "Hotel Pickup" add-on was selected (FR-067), optional otherwise.
FR-074	A **"Special Requests"** text area (500 char max) shall be available.
FR-075	A **"Promo Code"** input shall validate the code and apply the discount on-submit (FR-096).
FR-076	A T&C acceptance checkbox shall be mandatory.

9. Payment & Confirmation

9.1 Overview

Payment is the final friction point. Multi-gateway support is essential: Stripe for international cards, Safepay for local Pakistani methods. Multi-currency display is mandatory for conversion.

9.2 Related BRs

BR-014, BR-015, BR-016, BR-017

9.3 Currency & Pricing Display

FR ID	Requirement
FR-077	The system shall detect customer currency from IP geolocation on first visit (fallback: USD).
FR-078	A currency switcher in the header shall allow override (USD, GBP, PKR).
FR-079	Prices stored in PKR shall be converted via daily FX rate cached from a public API.
FR-080	At checkout, the final charge shall be in the gateway's processing currency, with FX conversion disclosed.

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9.4 Payment Gateway Routing

FR ID	Requirement
FR-081	For USD/GBP checkout, the customer shall be routed to Stripe Elements embedded on the checkout page.
FR-082	For PKR checkout, the customer shall be routed to Safepay (which wraps JazzCash, Easypaisa, debit/credit cards, bank transfer).
FR-083	If gateway initialisation fails, a fallback message shall offer the alternate gateway.
FR-084	Gateway webhook events shall be the source of truth for payment status (not the client callback).
FR-085	The booking record shall move from `pending_payment → confirmed` only on authenticated webhook of successful payment.

9.5 Confirmation

FR ID	Requirement
FR-086	On confirmation, the customer shall land on `/booking/[id]/success` showing booking summary, guide profile, itinerary, "Add to Calendar" (ICS) button, and WhatsApp CTA.
FR-087	A confirmation email shall be sent to the customer within 60 seconds (FR-199).
FR-088	A booking notification email shall be sent to the assigned guide within 60 seconds.
FR-089	A receipt PDF shall be generated and attached to the customer email.

9.6 Failed Payments

FR ID	Requirement
FR-090	A failed payment shall leave the booking in `pending_payment` state with slot held for 15 minutes.
FR-091	After 15 minutes without resolution, the slot shall be released.
FR-092	A reminder email shall be sent at 10 minutes if payment is incomplete.

10. Customer Dashboard

10.1 Overview

Logged-in customers manage their trips, review history, referrals, and account from a single hub.

10.2 Related BRs

BR-018

10.3 Dashboard Home

FR ID	Requirement
FR-093	The dashboard home (`/account`) shall display: upcoming bookings (card list), quick stats (total trips, upcoming count), featured content, referral CTA.

10.4 Bookings View

FR ID	Requirement
FR-094	A `/account/bookings` route shall list all bookings with tabs: **Upcoming**, **Past**, **Cancelled**.
FR-095	A booking detail page shall show full itinerary, guide WhatsApp CTA, receipt download, cancellation request (opens a mailto for v1 — automated refund in v2).

10.5 Reviews View

FR ID	Requirement
FR-096	A `/account/reviews` route shall list past bookings eligible for review and submitted reviews.

10.6 Referrals View

FR ID	Requirement
FR-097	A `/account/referrals` route shall display the unique referral link, number of successful referrals, credit earned, and a share-to-WhatsApp button.

11. Reviews & Ratings

11.1 Overview

Reviews are the brand's trust engine. The flow is simple: completed trip → prompt → star rating + text + optional photos → admin moderation → publish.

11.2 Related BRs

BR-004, BR-019

11.3 Review Eligibility

FR ID	Requirement
FR-098	A customer may submit a review only for a booking in state `completed`.
FR-099	A customer may submit exactly one review per booking.
FR-100	The review window shall be open from trip-complete until 30 days thereafter.

11.4 Review Submission

FR ID	Requirement
FR-101	The form shall capture: 1–5 star rating (required), review title (optional, 80 chars), review text (required, min 20 chars, max 2000), up to 5 photos (JPEG/PNG, max 5MB each).
FR-102	Photos shall be uploaded with client-side preview and server-side virus scan (ClamAV or equivalent).
FR-103	Submitted reviews enter `pending_moderation` state (FR-156).

11.5 Review Display

FR ID	Requirement
FR-104	Approved reviews shall display customer first name + last initial (e.g. "Sarah K."), country flag, rating, title, text, photos (lightbox), date, package reference.
FR-105	Reviews shall be sortable by Most Recent, Highest Rated, Lowest Rated.

11.6 Prompts

FR ID	Requirement
FR-106	24 hours after `completed` state, an email shall prompt the customer to review (FR-199).
FR-107	A reminder shall fire at 7 days if no review submitted.

12. Referral & Promo Codes

12.1 Overview

Growth loops: promo codes drive first-booking conversion; referrals drive loop acquisition.

12.2 Related BRs

BR-013, BR-020

12.3 Promo Codes

FR ID	Requirement
FR-108	A promo code input on checkout (FR-075) shall accept a code, validate server-side, and apply the discount.
FR-109	Codes shall support discount types: percentage, flat amount.
FR-110	Codes shall support constraints: minimum booking amount, max uses globally, max uses per customer, valid date range, applicable tiers.
FR-111	Invalid or expired codes shall return a clear error.
FR-112	Only one promo code may be applied per booking.

12.4 Referrals

FR ID	Requirement
FR-113	Each customer shall have a unique referral code generated at signup (e.g. `AHMAD-A7F2`).
FR-114	A referred customer applying the code at signup OR on first booking shall receive a signup discount.
FR-115	On the referred customer's first completed booking, the referrer shall earn account credit.
FR-116	Credit is held in the customer's "wallet" (visible on dashboard) and auto-applies to the next booking.

13. Safety Features (Universal, PWA-based)

13.1 Overview

Safety features apply to **every booking** regardless of tier or traveller profile. They are opt-in during the booking flow (FR-068), enabled by default when the customer has an emergency contact set. Guide check-ins are mandatory on every trip. This "universal safety layer" is the platform's primary trust differentiator in the Pakistani market.

In v1, safety is delivered through the **Progressive Web App (PWA)** — mobile browsers (iOS Safari, Android Chrome) support Web Geolocation API for GPS, service workers for background tasks, and Web Push for alerts. Customers are prompted to "Add to Home Screen" during booking so the PWA installs as an app-like icon. Phase 2's native Flutter app replaces/augments this with native OS capabilities.

13.2 Related BRs

BR-028, BR-029, BR-030, BR-031, BR-032, BR-033

13.3 Emergency Contact

FR ID	Requirement
FR-117	Any customer may set an emergency contact (FR-041); doing so unlocks the Safety Layer by default on their next booking.
FR-118	The emergency contact shall be stored encrypted at rest.

13.4 SOS

FR ID	Requirement
FR-119	During any active trip (state: `in_progress`) with Safety Layer opted-in, a prominent SOS button shall be available in the PWA (mobile browser) and on the customer dashboard.
FR-120	Triggering SOS shall immediately: (a) SMS the emergency contact with the customer's current GPS coordinates, (b) email the emergency contact, (c) alert the admin dashboard with a visible red flag, (d) log the event.
FR-121	A triggered SOS shall remain active until cancelled by the customer OR resolved by admin.

13.5 Live Trip Tracking

FR ID	Requirement
FR-122	During an active trip with Safety Layer opted-in, the PWA shall transmit the customer's GPS location every 60 seconds via the Web Geolocation API; service worker background sync queues updates if the browser is in the background.
FR-123	A share-link (expiring at trip end + 1 hour) shall be auto-sent to the emergency contact at trip start when the Safety Layer is active.
FR-124	The emergency contact shall view a web page (no login required) showing live location on a map.

13.6 Guide Check-ins (every trip)

FR ID	Requirement
FR-125	Assigned guides shall perform three mandatory check-ins per trip, regardless of Safety Layer status: (1) **Trip Start** (I'm on the way), (2) **Pickup Done** (customer is in my care), (3) **Trip Complete** (customer safely dropped off). Check-ins are completed via the guide view of the PWA on their phone browser.
FR-125a	For multi-day packages, check-ins recur per day: **Day Start**, **Day Complete**, plus the trip-wide **Trip Start** and **Trip Complete**.
FR-126	A missed check-in by 30 minutes past scheduled window shall raise an admin alert.

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13.7 Admin Safety View

FR ID	Requirement
FR-127	The admin panel shall display a real-time "Active Trips" board for all in-progress bookings with status, location (if Safety Layer active), check-in progress, and alert flags.

13.8 PWA Installability

FR ID	Requirement
FR-127a	The web app shall include a valid Web App Manifest (`manifest.webmanifest`) declaring name, icons (192×192, 512×512), theme color (hunter green), display mode `standalone` , start URL.
FR-127b	After a successful booking, the PWA shall prompt the customer to "Add to Home Screen" (iOS Safari and Android Chrome native prompt).
FR-127c	A service worker shall cache the shell (landing page, catalog, booking dashboard) for offline access.

14. Content & Static Pages

14.1 Overview

Content supports SEO and trust. Approved copy (brand/brand-content.md) is loaded; CMS updates in v1 are via static markdown.

14.2 Related BRs

BR-005

14.3 Pages

FR ID	Requirement
FR-128	A `/about` page shall render the approved "About Us" copy.
FR-129	A `/story` page shall render the Founder's Message and the Our Story narrative.
FR-130	A `/vision` page shall render Vision and Mission content.
FR-131	A `/safety-guide` page shall render customer-facing safety guidance (what to pack, what to expect, how SOS works).
FR-132	A `/terms` and `/privacy` page shall render legal copy (final text TBD by client's legal reviewer).
FR-133	A `/contact` page shall provide a contact form (name, email, subject, message) sending to a configurable admin inbox + a WhatsApp link.

Part III — Admin Panel

15. Admin Authentication & RBAC

15.1 Overview

Admin is a protected surface. Only invited users with assigned roles may access.

15.2 Related BRs

BR-034 (and NFR-018)

15.3 Roles

Role	Can Do
Super Admin	Everything, including role management
Ops Admin	Guides, Packages, Bookings, Payments, Reviews
Content Manager	Blog, Travel Guide, FAQ, Static Pages, Newsletter

15.4 Requirements

FR ID	Requirement
FR-134	Admin signup shall be by invitation only (super admin generates invite link).
FR-135	Admin login shall require email + password + optional 2FA (TOTP).
FR-136	After 3 failed login attempts, account locked for 30 minutes.
FR-137	Admin session shall expire after 60 minutes of inactivity.
FR-138	Every admin action shall be logged to an audit trail (who, what, when, before/after).
FR-139	The super admin shall be able to assign/change roles.

16. Guide Management

16.1 Related BRs

BR-021, BR-022, BR-023, BR-024, BR-025, BR-027

16.2 Guide List

FR ID	Requirement
FR-140	A table at `/admin/guides` shall list guides with: avatar, name, status (pending / verified / suspended), cities, tiers, rating, booking count.
FR-141	Filters: status, city, tier, gender tag.

16.3 Guide Profile CRUD

FR ID	Requirement
FR-142	Admin shall create/edit a guide with: full name, gender tag (female/male/other, required), photo, bio, languages (multi-select), cities (multi-select), tiers served (multi-select), specialities, WhatsApp number, flat fee per booking, availability default pattern.
FR-143	Admin shall upload: CNIC image (front + back, encrypted), police clearance document, references document.
FR-144	Admin shall move a guide through states: `pending → verified → active → suspended / retired`.
FR-145	Only `active + verified` guides shall be eligible for new assignments (FR-022).

16.4 Availability Calendar

FR ID	Requirement
FR-146	Each guide profile shall have an availability calendar — days marked available/unavailable.
FR-147	Admin can bulk-mark days (e.g. "all Mondays unavailable for next 3 months").

17. Package & Tier Management

17.1 Related BRs

BR-037, BR-038

17.2 Tier Management

FR ID	Requirement
FR-148	Admin shall CRUD tiers with: name, slug, description, cover icon, suggested-speciality tags, display order.

17.3 Package CRUD

FR ID	Requirement
FR-149	Admin shall CRUD packages with: title, slug, tier, city, duration (hours), base price (PKR), included services (list), excluded services (list), itinerary steps (ordered), eligible guides (filter by tier/city), status (draft/published/archived), featured flag, cover + gallery images, SEO fields.
FR-150	Changing a package's tier shall require confirmation and re-filter eligible guides.

17.4 Add-ons

FR ID	Requirement
FR-151	Admin shall CRUD add-ons: name, description, price (PKR), image, applicable tier(s).

17.5 Excel/Doc Importer

FR ID	Requirement
FR-152	Admin shall upload an Excel (`.xlsx`) or Word (`.docx`) file conforming to a published template.
FR-153	The system shall parse the file and present a preview table of packages to be created/updated.
FR-154	On confirm, packages shall be created in `draft` state; admin publishes manually.
FR-155	Validation errors shall be reported per row with inline fix capability.

18. Booking Management

18.1 Related BRs

BR-036, BR-039

18.2 Requirements

FR ID	Requirement
FR-156	A table at `/admin/bookings` shall list all bookings with: ID, customer, package, date, tier, guide, status, payment status, total.
FR-157	Filters: status, tier, date range, guide, city.
FR-158	A booking detail page shall show full data + action buttons: reassign guide, mark complete, cancel, refund (manual).
FR-159	Reassigning a guide shall notify both the removed and the newly assigned guide via email.
FR-160	Admin shall add internal notes to a booking (not visible to customer).

19. Payment & Refund Management

19.1 Related BRs

BR-040

19.2 Requirements

FR ID	Requirement
FR-161	A table at `/admin/payments` shall list all transactions with: date, booking ID, customer, gateway, currency, amount, status, fee, net.
FR-162	Reconciliation: admin shall see payment total vs. gateway payout total per period.
FR-163	A refund flow shall call the gateway's refund API; status tracked on the booking.
FR-164	A **Guide Payout Report** at `/admin/payouts` shall list each guide's completed bookings × flat fee for a chosen period, with mark-as-paid action.

20. Review Moderation

20.1 Related BRs

BR-041

20.2 Requirements

FR ID	Requirement
FR-165	A table at `/admin/reviews` shall list reviews with status (`pending` / `approved` / `rejected` / `hidden`).
FR-166	Each review's detail view shall show full text, photos, customer, package, guide, booking context.
FR-167	Admin may approve, reject (with reason, emailed to customer), or hide (previously approved review).
FR-168	Rejected reviews are retained for 90 days then purged.

21. Content Management System (CMS)

21.1 Related BRs

BR-005

21.2 Requirements

FR ID	Requirement
FR-169	A `/admin/blog` CMS shall allow create/edit/publish of blog posts with rich-text editor (Tiptap or similar), cover image, tags, category, status (draft/published/scheduled), SEO fields.
FR-170	A `/admin/travel-guide` CMS shall allow the same for travel guide articles.
FR-171	A `/admin/faq` CMS shall allow CRUD of FAQ items grouped into categories.
FR-172	A `/admin/pages` CMS shall allow editing of static pages (About, Vision, Our Story, Safety Guide) via markdown or rich text.
FR-173	A "**Preview**" button shall render the page as it will appear publicly before publishing.

22. Promo Code Management

22.1 Related BRs

BR-042

22.2 Requirements

FR ID	Requirement
FR-174	A `/admin/promo-codes` table shall list codes with: code, type, value, usage count, validity, status.
FR-175	Admin shall create a code (constraints per FR-110), edit, deactivate.
FR-176	Clicking a code shall show its usage log (who used, when, booking ID).

23. Newsletter Management

23.1 Related BRs

BR-045

23.2 Requirements

FR ID	Requirement
FR-177	A `/admin/newsletter` page shall list subscribers (email, country, signup date, status).
FR-178	Admin shall export subscribers to CSV.
FR-179	Admin shall mark a subscriber as unsubscribed (manually) or honour unsubscribe requests from email links.
FR-180	*(v2)* Admin shall compose and send newsletter campaigns from within the panel — v1 uses an external tool (Mailchimp/Resend) with list sync.

24. Analytics & Reporting

24.1 Related BRs

BR-044

24.2 Requirements

FR ID	Requirement
FR-181	A `/admin/analytics` dashboard shall display, for a selectable period: bookings count, revenue (PKR), average rating, top packages, tier mix, guide utilisation leaderboard, new signups, newsletter signups.
FR-182	Each metric shall be downloadable as CSV.
FR-183	The dashboard shall default to the current month vs. previous month.

Part IV — API & Mobile Interface

25. API Overview & Conventions

25.1 Requirements

FR ID	Requirement
FR-184	The API shall be versioned at `/api/v1/*`.
FR-185	All endpoints shall return JSON with a consistent envelope: `{ data, error, meta }`.
FR-186	OpenAPI 3.1 spec shall be auto-generated and hosted at `/api/docs` (admin-restricted in prod).
FR-187	Authentication for the API shall be JWT access + refresh (FR-047).
FR-188	Rate limiting: 100 req/min per IP for public; 500 req/min for authenticated.
FR-189	All timestamps shall be ISO 8601 UTC.
FR-190	Pagination shall use `?page=N&per_page=M` with response meta `{ total, page, per_page, pages }`.
FR-191	Webhooks (payment, SMS delivery) shall be verified by HMAC signature; replays rejected via nonce.

26. Native Mobile App (Phase 2 — post-v1 launch)

26.1 Related BRs

BR-046, BR-049 (deferred from v1 to Phase 2)

26.2 Scope Note

The native Flutter mobile app is a Phase 2 deliverable (post-launch, weeks 13–20). In v1, the PWA (§13.8) delivers equivalent capabilities on mobile browsers. The requirements below are specified now for

forward-compatibility and Flutter-dev onboarding; they are NOT built in v1.

26.3 Requirements (Phase 2)

FR ID	Requirement
FR-192	The native app shall support browsing, booking, payment, dashboard, reviews — feature parity with the PWA for the customer flow, with native UX polish (gestures, animations, platform navigation).
FR-193	The native app shall provide the SOS button and live location tracking as described in §13, using OS-native location services (more battery-efficient and reliable in background than Web Geolocation).
FR-194	The guide view in the native app shall allow: view assigned bookings, perform the 3 check-ins (FR-125, FR-125a), update own availability, view earnings.
FR-195	FCM push notifications shall be sent on: booking confirmation (customer), booking assignment (guide), SOS trigger (admin), check-in missed (admin), review published (customer). Replaces the v1 Web Push baseline.
FR-196	The native app shall cache the catalog + bookings + in-flight check-in data for full offline operation.
FR-197a	The native app shall support a native photo-capture flow for reviews, including auto-rotation, cropping, and compression.

Part V — Cross-Cutting Concerns

27. Notifications

27.1 Email

FR ID	Requirement
FR-197	Email delivery shall use Resend as transactional email provider.
FR-198	Templates (HTML + plain text) shall be versioned in-repo and brand-styled (hunter green + gold).
FR-199	Transactional emails include: welcome, email verification, password reset, booking confirmation (customer), booking assignment (guide), payment receipt, review prompt, review published, SOS alert, admin role invitation.
FR-200	All emails shall include an unsubscribe footer for marketing categories only; transactional emails are exempt but allow preference adjustment.

27.2 SMS

FR ID	Requirement
FR-201	SMS delivery shall use Twilio with PK and international numbers.
FR-202	SMS use cases: SOS alert (emergency contact), OTP (future), guide late-check-in warnings.

27.3 Push Notifications

FR ID	Requirement
FR-203a	**v1:** Web Push via service worker (`PushManager` + VAPID keys). Customer opts in on first visit after signing up.
FR-203b	**Phase 2:** Native push via Firebase Cloud Messaging (FCM) replaces/augments Web Push on native app.
FR-204	Users shall manage push preferences in their account settings.

27.4 WhatsApp

FR ID	Requirement
FR-205	WhatsApp click-to-chat links (`wa.me/<number>?text=<prefilled>`) shall surface post-booking on booking detail + confirmation email.
FR-206	Guide WhatsApp numbers shall NOT be shown pre-booking.

28. Third-Party Integrations

FR ID	Integration	Purpose
FR-207	Stripe	International card payments (USD, GBP)
FR-208	Safepay	Local PK payments (JazzCash, Easypaisa, cards)
FR-209	Resend	Transactional email
FR-210	Twilio	SMS
FR-211a	Web Push (VAPID)	v1 push notifications via service worker
FR-211b	FCM	Phase 2 native push notifications
FR-212	Google OAuth	Social login
FR-213	Facebook OAuth	Social login
FR-214	Apple Sign-In	Mobile social login
FR-215	Google Maps API	Location display, SOS tracking
FR-216	OpenExchangeRates (or similar)	Daily FX rates
FR-217	Sentry	Error tracking
FR-218	PostHog or GA4	Product analytics

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29. Error Handling & Validation

29.1 Requirements

FR ID	Requirement
FR-219	All form inputs shall be validated client-side (immediate feedback) AND server-side (authoritative).
FR-220	API error responses shall use HTTP status codes correctly (400 validation, 401 unauth, 403 forbidden, 404 not found, 409 conflict, 422 unprocessable, 500 server).
FR-221	Error response body shall include: `error.code` (machine-readable), `error.message` (human-readable), `error.fields` (field-specific errors).
FR-222	Server errors (5xx) shall be logged to Sentry with context.
FR-223	The user-facing web shall display friendly error messages (never raw stack traces).
FR-224	A 404 page shall offer navigation back to home + catalog.
FR-225	A 500 page shall apologise and offer a WhatsApp/contact fallback.

Part VI — Appendices

30. Appendix A — BR → FR Traceability Matrix

BR	Addressed By
BR-001	FR-001–FR-004 (SEO + landing)
BR-002	FR-002, FR-003
BR-003	FR-025–FR-029 (newsletter)
BR-004	FR-008, FR-104–FR-105
BR-005	FR-020–FR-024, §21 CMS
BR-006	FR-012–FR-019
BR-007	FR-049–FR-058 (package detail)
BR-008	FR-071
BR-009	FR-030, FR-036
BR-010	FR-059–FR-062
BR-011	FR-063–FR-066
BR-012	FR-067–FR-069
BR-013	FR-075, FR-108–FR-112
BR-014	FR-077–FR-081
BR-015	FR-082
BR-016	FR-086–FR-089
BR-017	FR-094–FR-095
BR-018	FR-093–FR-097
BR-019	FR-098–FR-107
BR-020	FR-113–FR-116
BR-021	FR-144–FR-145

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BR-022	FR-142–FR-144
BR-023	FR-054
BR-024	FR-146–FR-147
BR-025	FR-088
BR-026	FR-164
BR-027	FR-054, FR-104
BR-028	FR-041, FR-117–FR-118
BR-029	FR-119–FR-121
BR-030	FR-065
BR-031	FR-125–FR-126
BR-032	FR-122–FR-124
BR-033	FR-127
BR-034	§15–§23 (all admin modules)
BR-035	FR-152–FR-155
BR-036	FR-158–FR-159
BR-037	§17
BR-038	FR-166–FR-167
BR-039	FR-174–FR-176
BR-040	FR-164
BR-041	§20
BR-042	§22
BR-043	FR-164
BR-044	FR-181–FR-183
BR-045	FR-177–FR-180
BR-046	FR-192–FR-196
BR-047	FR-118, §29 (compliance)
BR-048	FR-138, FR-161

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BR-049	FR-195, FR-203
BR-050	Full data model forward-compatibility (addressed in SRS)

31. Appendix B — Open Items

Items awaiting resolution before FRD v1.0 approval:

1. Final tier naming (Women-First / Cultural Heritage / Food Trail / Family / Nature) — confirm from Aiman
2. Lahore activity catalog — feeds the actual package list and itineraries
3. Guide roster with gender tags + languages + specialities
4. Terms & Privacy legal copy (client's legal reviewer)
5. Cancellation & refund policy wording (deferred per BRD D-004)
6. Default availability calendar pattern for guides
7. Analytics tool choice confirmation (PostHog vs GA4)
8. Logo variations needed: dark background, square avatar, favicon sizes
9. OTP delivery choice (SMS now vs later)
10. Whether `/admin/newsletter` v1 integrates with Mailchimp vs Resend Audiences

32. Appendix C — Change Log

Version	Date	Author	Summary
v0.1	2026-10-19	Ahmad Fawad, Flyxosoft	Initial FRD translating BRD v0.2 business requirements into 225 numbered functional requirements across customer web, admin panel, API/mobile, and cross-cutting concerns. BR → FR traceability matrix included.
v0.2	2026-10-19	Ahmad Fawad, Flyxosoft	**Round 4 refinements.** (1) **Women-First tier removed.** Safety features (SOS, live tracking, emergency contact, guide check-ins) made universal across all bookings — §13 renamed "Safety Features (Universal)". Concierge add-ons made universal optional add-ons at checkout for any package. FR-065 reframed as a gender preference filter. FR-067-FR-069 universalised. FR-072 tied to Safety Layer opt-in. (2) Aiman, Umaimah, Waqar Ullah listed as Co-Founders (equal) throughout. (3) Tier list updated (removed Women-First from references). (4) BR-030 maps to new gender-preference FR instead of tier-scoping.
v0.3	2026-10-19	Ahmad Fawad, Flyxosoft	**Ahmad Fawad added as the 4th Co-Founder** of Daastaan-e-Pakistan (dual role — also Flyxosoft PM / Lead). Reflected in Document Control "Prepared For".
v0.4	**2026-10-19**	**Ahmad Fawad, Flyxosoft**	**Round 5 scope expansion.** (1) **Multi-city, multi-day, group bookings pulled into v1** — new/updated FRs: FR-012 (city picker), FR-012a/b (per-city SEO + master catalog), FR-013 (city/duration/group filters), FR-050 (duration + group range on card), FR-051 (multi-day itinerary grouping), FR-059 (180-day window), FR-061a/b/c/d (date range, party size, group pricing). (2) **Native Flutter mobile → Phase 2** — §26 re-scoped as Phase 2 deliverable. (3) **v1 web is a PWA** — §13 re-anchored on Web Geolocation + service worker; new §13.8 PWA installability FRs (FR-127a/b/c); FR-125a added for multi-day check-ins. (4) Push notifications split into v1 Web Push (FR-203a) + Phase 2 FCM (FR-203b). (5) §2 Scope restructured into v1 / Phase 2 / v2+ with new §2.4 PWA rationale.

End of FRD v0.4 Draft

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